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SWE Project Proposal Fall 2026 - Klepetko

Overall Goals

- Create a video game with a rival that adapts to compete against you, the player. The chosen format is that of a card game; Specifically a roguelike deck-builder inspired by Slay the Spire.

Scope Boundary

- Increasingly difficult opponents for the player to face as they progress through the endless ladder in addition to their Rival.
- After each match, the player will be rewarded with new and often stronger cards, and be given an opportunity to edit their deck by adding or removing cards they own.
- A simple opponent AI that utilizes the current deck to the best of its ability to defeat the player. This AI is used to determine the cards played by both the Rival AND the other opponents.
- A complex Deck-Building AI which constructs its own deck based on what the player had when they fought them last. The AI will select cards with the goal of constructing a deck that can defeat the player's previous deck.
- Development of a basic art style/direction for the game. Doesn't need to be final or polished, but gameplay needs to be communicated effectively.

Project Features

- Mobile and web friendly card strategy video game
- Procedural cards, Deck-building
- Rival AI that learns from human player's deck and play style

Project Requirements

- HTML5 game engine that handles events and the main render loop
- Game interfaces with custom learning AI agent (runs on device, traditional non-generative machine learning)
- CardGameState Interface that can be used for creating unique card behaviors
- Basic preset AI plays appropriate available cards based on the current match state
- Procedural card library generation

Task Breakdown

- Development Tools Setup
 - Github repository
 - branching and pull request convention
 - internal documentation/guides
- Engine Programming
 - HTML5 game that we can embed in any site easily
 - asset format/loading
 - game states:
 - main menus,
 - in round,
 - between rounds
 - map/quest navigation, levels
 - shop
 - deck build
- Card Play System
 - health
 - energy
 - deck
 - hand
 - cards currently in play
- Procedural Card System
- UI Design/Layout
- AI System Development
- AI Tuning
- Mechanics Programming
- Prose Writing/Dialog
- Image Asset Sourcing/Creation
 - Card backgrounds/art
 - Playing field backgrounds/art
 - Characters
 - Particle fx
 - UI Design
- Audio Asset Sourcing/Creation

Delegation of Responsibilities

- Jack: Dev tooling, repository management, engine programming, audio, testing/qa, code inspection/documentation
- Chris J: Debug, playtest, feature integration, deck-building AI
- Chris B: Procedural generation of playable card data, gameplay balance, card-playing AI
- Julia: User Interface, User Experience, graphic design, team coordination

Timeline

- Phase 1: Create playable game with dummy AI
- Phase 2: Implement learning AI system
- Phase 3: Polish the game and the integration with the AI system